



Research article

Re-establishment of hummock topography promotes tree regeneration on highly disturbed moderate-rich fens

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ABSTRACT

Winter exploration of oil sands deposits underlying wooded fens mostly eliminates the hummock-hollow topography on drilling pads and the ice roads leading to them, after their abandonment in spring. Recovery of black spruce (*Picea mariana* (P. Mill.) B.S.P.) and tamarack (*Larix laricina* (Du Roi) K. Koch) on these disturbed peatlands is thought to depend on the recovery of hummock topography. In late winter, numerous large blocks of frozen peat (1.5 × 1.5 m) were lifted out of the flattened drilling pads and positioned beside their excavated hollows; this was done on six temporary pads. Four years later, the condition of the mounds and the regeneration of conifers from natural seed dispersal were assessed on these elevated mounds compared to adjacent flattened areas of the pads. Then, conifer seedling density was more than five times higher on elevated spots than the mostly flat, flood-prone areas between them, and seedling density was positively related to mound height and strength of seed source. Higher mounds tended to have larger seedlings. Mounds on some of the pads were heavily eroded down; these pads had peat with higher humification, and operationally these pads were also treated in late winter when peat was thawing and fractured into pieces during mound construction. Developing a large volume of elevated substrate that persists until natural hummock-forming mosses can establish is thought necessary for tree recruitment and the recovery of the habitat for the threatened woodland caribou of this region.

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1. Introduction

Canada has 1.24×10^6 km² of peatlands and much of the northern half of Alberta is covered by peatland (Vitt et al., 1997). In many regions, fens blanket half of the landscape and have developed over the last 7000 years from pond loci and then by paludification and lateral expansion (Halsey et al., 2000; Bauer et al., 2003). Wooded moderate-rich (WMR) fens are the most common peatland type in this area and are characterized in part by hummock-hollow microtopography (Vitt, 1994), where hummocks, formed by mosses such as *Sphagnum fuscum* (Schimp.) Klinggr., are 30–50 cm above the usual water table and the hollows usually have standing water. These fens have a tree layer that is dominated by black spruce (*Picea mariana* (P. Mill.) B.S.P.) and tamarack (*Larix laricina* (Du Roi) K. Koch) and their roots are confined to positions above the usual water table (Lieffers and Rothwell, 1987).

Alberta has large tracts of deeply buried bitumen reserves that underlie 142,000 km² of Alberta (Government of Alberta, 2016). The exploration of the oil sands deposits by *in situ* methods places oils sands exploration (OSE) drilling pads at a density of up to five pads per km², even across wooded fens. In exploration areas across peatlands, a network of temporary OSE ice drilling pads, and associated winter ice roads for access, are established (see details below). Caners and Lieffers (2014) surveyed 10-year old abandoned OSE pads that were formed by this freezing-in processes and showed that most pads lose the hummock-hollow microtopography typical of WMR fens. The highest surfaces on most abandoned pads are usually within 8 cm of the water table and the dominant vegetation is aquatic sedges. As roots of northern conifers such as black spruce and tamarack cannot tolerate extended periods of anaerobic conditions (Conlin and Lieffers, 1993), trees germinating close to the water table height will experience periods of anaerobic rooting conditions. Thus, restoration treatments to produce acceptable growing sites for conifer seedlings on such heavily disturbed peatland pads will likely need substrate positions to be well above the water level (Pearson et al., 2011). This needs

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testing.

In general, most of the literature on the recovery of fen ecosystems after disturbance has focussed on re-establishment of peatland vegetation following drainage for forestry (Laine et al., 2011) or drainage plus peat removal and its resulting shifts in peatland chemistry and prolonged loss of peatland vegetation (see reviews by Grand-Clement et al., 2015; Lamers et al., 2015). Here the application of donor plant diaspores is often used to establish peatland mosses and other vegetation on the base substrates in mined-out areas (Graf and Rochefort, 2008; Pouliot et al., 2011). Re-development of microtopography has been identified (Triisberg et al., 2014) as important in restoration of peatland vegetation. Recovery of frozen-in OSE pads and roads on fens, however, has a different starting point than most other disturbed fens; lack of microtopography and flooding might be the critical factors for species re-establishment, instead of lack of propagules and changes in peatland chemistry in highly disturbed drained fens.

The woody vegetation on these WMR fens provides key habitat characteristics for the threatened Woodland Caribou (*Rangifer tarandus caribou*) of this region (Species at Risk Act, Schedule 1; Environment Canada, 2012). Coniferous trees are thought to slow the movements of people (Pigeon et al., 2016) and predators into these habitats, and trees provide substrate for arboreal lichens, a food source for caribou (Environment Canada, 2012); thus trees are a focus of recovery efforts for OSE pads and the thousands of kilometers of seismic exploration lines across peatlands (van Rensen et al., 2015). Recovery of trees on such disturbed fens, however, is slow (Lee and Boutin, 2006; van Rensen et al., 2015). The natural recovery of conifers on peatlands is promoted by minor disturbance to substrates and more elevated microsites (Morris et al., 2009) and the amount of seed cast (Groot and Adams, 2005). Seed of tamarack and spruce may readily spread 70 m into disturbed areas from living trees on the edges (Greene and Johnson, 2000), but it is not clear if this would be the case with the smaller trees of peatlands. As these peatlands are remote and access for planters during summer months can be treacherous, natural regeneration as opposed to planted trees would be desirable.

In this study, we examined the effectiveness of artificial mound formation to promote the natural regeneration of trees on highly disturbed fens. Assessments were made four to five years after mounding, on six OSE drilling pads on WMR fens. We linked regeneration success to the height of mounds, the density and size of adjacent seed trees and peat characteristics among pads. As peat surfaces without living mosses and vascular plants are subject to erosion (Parry et al., 2014; Shuttleworth et al., 2015), we also assessed if artificial mounds were sustained over time.

2. Methods

This study was conducted in a large fen complex, ca. 40 km southeast of Conklin, Alberta (central study drilling pad 2-32: 55° 26' 52.3" N, 110° 53' 23.9" W), within the Central Mixedwood Natural Subregion of the Boreal Forest Natural Region (Natural Regions Committee, 2006). This peatland complex is characterized by aeolian sand outcrops, and variation in peatland type (Grootjans et al., 2012); elevational changes and water flow patterns are easily visible from aerial and global mapping (see attachment). Sites were categorized as wooded moderate-rich fens based upon surface water pH 5.5 to 7, moderate alkalinity (Table 1) and indicator bryophytes, including minerotrophic *Sphagnum* species, *Tomentypnum nitens* (Hedw.) Loeske, *Hamatocaulis vernicosus* (Mitt.) Hedenas, and *Brachythecium mildeanum* (Schimp.) Schimp. ex Milde (Vitt, 1994; National Wetlands Working Group, 1997). Climate is continental: mean daily temperature is 2.1 °C and mean total annual precipitation is 421 mm (Environment

Canada, Cold Lake A meteorological station; http://climate.weather.gc.ca/climate_normals/index_e.html).

We examined six temporary drilling pads (up to 10 km apart, each approximately 0.7 ha) that were constructed by Devon Energy Corporation in winter 2011/2012 in their regular operations to assess the depth and quality of bitumen reserves. Starting in early winter, trees were cut and either mulched, or pushed (windrowed) to the side of the pad. Using freezing-in techniques, the snow and remaining vegetation including tree stumps and most of the root systems were smoothed, compacted and some had water added to allow greater penetration of frost into wet peat. The pads, and roads leading to them, became strong enough to support heavy machinery and drilling rigs; rigs were removed soon after drilling.

After rig removal, the frozen surface peat was sliced 70 cm deep in a grid pattern (1.5 × 1.5 m squares) with a single tooth shank ripper mounted on a bulldozer (D6 Caterpillar). A tracked excavator (Komatsu PC200LC) fitted with a standard bucket and thumb lifted out the frozen squares (35–60 cm thick) and repositioned these on the surface adjacent to their excavated holes with the original upper peat layer upright (Fig. 1a). One of the pads (6–5; numbers relate to a contraction of the mapping description) was mounded on February 4, 2012. In this colder period, large blocks of peat were lifted out intact (Fig. 1a). The remainder of the pads were mounded in early to mid March (Table 1). At this later time, there was some thawing and peat usually fractured into pieces during mound formation, except at 8–28 (Appendix 1). The mounds were left to regenerate naturally from tree seed dispersed by wind from adjacent stands (Fig. 1).

In either the fourth or fifth growing season, we sampled 10 mounds selected systematically across each of the six pads. Circular plots with a radius of 1.78 m (10 m²) centred on mounds; the 10 m² plot size is used in the regulatory surveys used for assessing regeneration. Near each mounded plot a second non-mounded plot was established in the adjacent area. In each of these 20 plots per pad, the number of seedlings of each tree species was counted and the height of tallest seedlings of each tree species. At each pad, peat samples were collected from 10 cm below the upper surface near the edge of four mounds. The degree of decomposition/humification of each peat sample was assessed using a visual analysis of the peat water and the textural state of the plant material forming the peat. This information was used to scale humification (H) using the von Post classification, graded from 1 to 10 (Hobbs, 1986), where H₁ meant that the moss and plant material was mostly intact and H₁₀ it was highly decomposed. We visually assessed the average height of the canopy conifer trees on four edges of the pad using a Biltmore stick and the density of seed trees in a 20 m ring around the pad was estimated visually by tree counts from 10 × 10 m virtual plots on aerial photographs (scaled by known dimensions). Using these quick estimates we ranked pads in terms of their potential seed supply; the mean rank of tree density and the mean height of the trees were averaged to obtain pad rank – analogous to strength of seed source assessed by basal area by Greene and Johnson (2000). Seed crops of tamarack and black spruce appeared to be small to moderate in the years from 2012 to 2015. In each mounded plot, the height of the mound above the substrate of the average surrounding area was measured using a horizontal bar and vertical ruler, set perpendicular to each other by a hand level. Pads 6–5 and 3–33 were assessed in late July of 2015, and pads 8–28, 7–32, 2–32 and 15–29 in late June of 2016. To standardize the presentation of seedling height, in 2016 seedlings were measured at the bud scale scar formed in 2015, i.e., at four years growth.

Water samples were obtained from these six sites, collected from natural pools near the centre of pads in 1 L Nalgene bottles. These were refrigerated within a few hours of collection and then frozen prior to analysis by the Natural Resources Analytical Laboratory (NRAL) at the University of Alberta, for pH, electrical

Table 1
Characteristics of drilling pads on a wooded moderate-rich fens near Conklin Alberta. Sites are ranked based upon the mean height of the mounds.

Site	¹ Mound ht cm	² Date	³ H	⁴ Spruce dens (ht)	⁴ Tamarack dens (ht)	⁵ Rank vigor	⁶ pH	⁶ EC $\mu\text{s cm}^{-1}$	⁶ Alkal. mg L^{-1}
8-28	39.6 (1.7)	76	2	2000 (8)	2000 (8)	2	6.9	203	99.6
6-5	37.5 (2.0)	35	4	2000 (10)	2000 (9)	1	7.3	250	147.1
3-33	22.2 (4.2)	76	4	1500 (7)	1500 (8)	3	7.1	243	137.1
7-32	20.7 (1.5)	67	3	200 (8)	1500 (6)	4.5	6.8	84	45.2
2-32	17.2 (1.6)	70	5	400 (5)	1500 (6)	4.5	6.5	74	44.5
15-29	8.9 (0.8)	75	6	100 (5)	1500 (5)	6	6.7	137	81.9

Notes
¹ Mean mound height (cm) and (s.e.) four summers after pad preparation.
² Annual day of pad completion in 2012.
³ Von Post peat humification (median of four samples per pad).
⁴ Adjacent tree density (trees ha⁻¹) and height (m) of co-dominants.
⁵ Mean site rank of conifer seed supply based upon the average rank of tree density and rank of average height of adjacent trees.
⁶ Water pH, electrical conductivity and alkalinity (mg L⁻¹ CaCO₃).

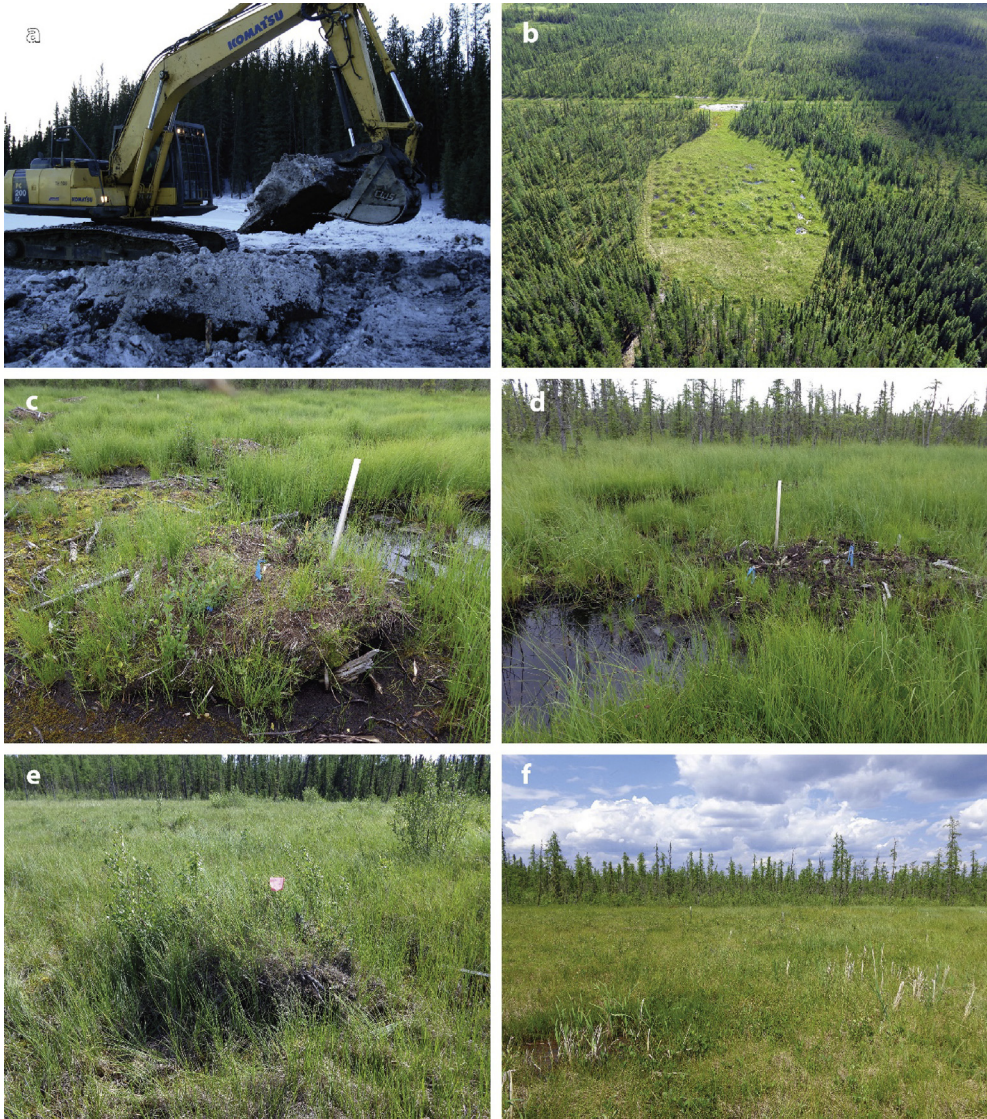


Fig. 1. Photographs of drilling pads at selected study sites. a) An excavator constructing mounds at pad 6-5, February 2012; peat frost was thick and strong at this time. b) Aerial photo of pad 6-5, late June 2016; note the persistent mounds and vigorous adjacent wooded peatland. c) Persistent mounds of pad 6-5 on July of 2013; note the sedge colonization and blocky structure but edge erosion of peat. d) Low, eroded mound on pad 2-32 on July of 2013. e) Intact mounds of pad 8-28, late June 2016. f) Low, eroded mounds of pad 15-29, late June 2016.

conductivity (μS) and alkalinity. Analytical techniques used are available at <http://nral.ualberta.ca/Analytical-Methods>.
We used paired t-tests to assess differences in tree seedling density between mounded and un-mounded plots of each pad.

Given only the moderate range in heights of mounds within OSE pads, however, we also decided to use regression techniques to analyze responses in relation to mound height across all the pads. For each test, variables were tested for normality and

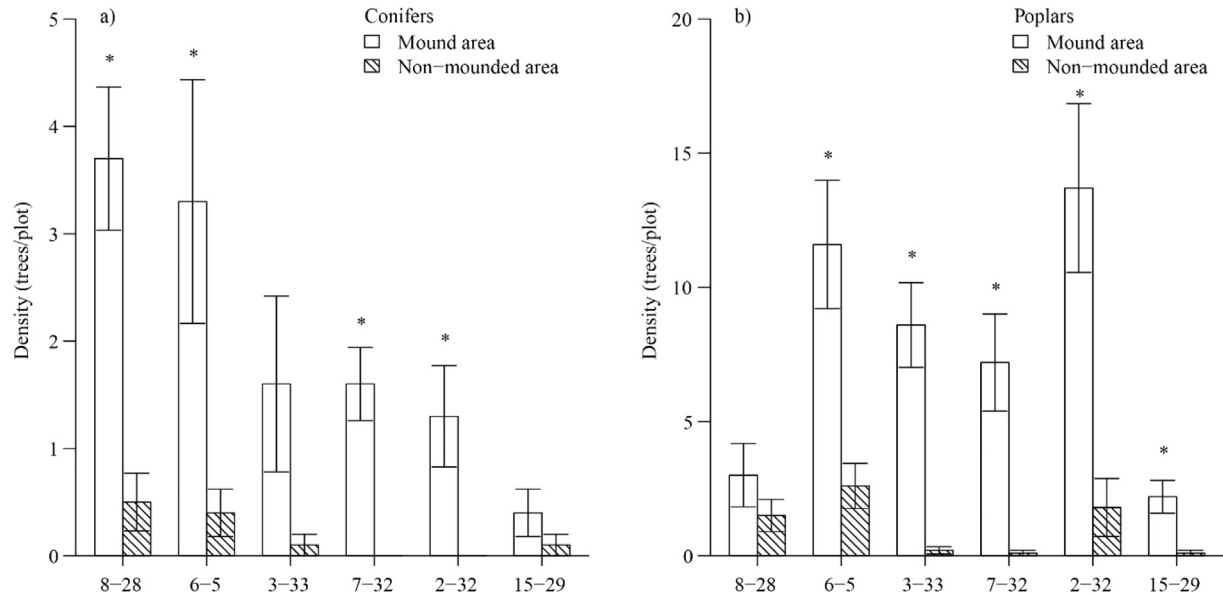


Fig. 2. Tree seedling density per 10 m² plot on mounded and non-mounded areas of each drilling pad (numbers along the x-axis). Values are mean $\pm$ 1 s.e.; * indicates P-values < 0.05 based on paired t-test, n = 10, for a) conifer species and b) poplar species. Pads are ranked from highest to lowest on the x-axis on the basis of mean height of mounds (see Table 1).

homoscedasticity. Tests had P values > 0.05.

3. Results

For all pads, the mounds had much greater density of conifer seedlings (Fig. 2a) than in adjacent non-mounded areas. First, there were no black spruce seedlings in non-mounded plots across the 6 pads and two of the pads had no tamarack seedlings in non-mounded plots. Across all pads, there were an average of 2.0 conifer seedlings in mounded plots and 0.2 seedlings in the non-mounded plots. Density of poplar seedlings was significantly greater in the mounded plots for five of the six pads (Fig. 2b, paired t-tests). Overall, there were 5.7 aspen (*Populus tremuloides* Michx.) and 2.0 balsam poplar (*Populus balsamifera* L.) in mounded plots, compared to 0.6 aspen seedlings and 0.5 balsam poplar in the non-mounded plots.

As there were considerable differences in tree density across the pads (Fig. 2), we tried to relate this to differences in environmental variables by ranking the pads by key environmental variables (Table 1). We first considered the height of the mounds at year four or five. Average density of conifer seedlings on mounds increased in a nearly linear pattern with average mound height ($r = 0.990$, $P < 0.001$, $n = 6$) (Fig. 2, note the ranking of the pads by mound height in the figure). The density of conifer seedlings in each individual plot also increased with mound height for overall plot averages (Fig. 3a, $r = 0.577$, $n = 60$, $P < 0.001$) and for two of the six individual pads there was significant positive relationship ($n = 10$, Appendix 2). The relationship between poplar seedling density vs. mound height was weaker on the basis of pad averages (Fig. 2b), but three of the six individual pads had significant positive slopes (Appendix 1). The mean frequency of mounds with a conifer seedling was 71.7%; i.e., more than five times that of the non-mounded areas of the pads (13.3%) (Appendix 3 and 2).

The height of the tallest black spruce or tamarack in the individual plots was weakly related to mound height (Fig. 4a) and there were no significant correlations between the heights of the tallest seedlings vs. mound height when testing mounds within each pad. The mean height of largest tamarack on individual mounds, across all six pads was 24.8 cm by year 4. The mean height of dominant black

spruce on mounds was 8.1 cm. The largest seedling in mounded plots increased with mound height for both aspen ($r = 0.611$, $P < 0.001$) and balsam poplar ($r = 0.529$, $P = 0.003$) (Fig. 3b) and one pad of the six had significant relationships within pads for aspen and balsam poplar. The mean height of dominant aspen was 33.3 cm on mounded plots; for balsam poplar this was 42.0 cm.

Mean mound height was also correlated with other environmental variables among the pads (Table 1). First, the rank of the vigor of the adjacent conifer stands was nearly in the same rank as the height of the mounds. Secondly, the von Post humification status was least decomposed on the pad with the highest mounds (8–28) and the two pads with the most humified peat had the lowest mounds (2–32 and 15–29). Qualitatively, the pads with poorest regeneration (2–32 and 15–29) were also the wettest and peat on the non-mounded surface of the pads was soft and floating at year five.

4. Discussion

This study shows that four or more years after abandonment of frozen-in well pads there was more than a five-fold increase in the density of natural regeneration of tamarack and black spruce on artificial mounds as compared to the more level and near-flooded areas without mounds. No black spruce seedlings were found in un-mounded areas. The artificial mounds created elevated locations where roots of the black spruce and tamarack could grow in substrates above the water table; such sites are necessary for survival of tamarack or spruce roots (Lieffers and Rothwell, 1987) and pine (Pearson et al., 2011) in peatlands. The elevated mounds are analogous to the hummock tops formed naturally by hummock-forming *Sphagnum* mosses such as *S. fuscum* (Vitt, 1994) in wooded poor- and moderate-rich fens. Recreating hummock microtopography was suggested for promotion of a variety of plant species (Triisberg et al., 2014) and may be the treatment needed for colonization of hummock-forming *Sphagnum* (Caners and Lieffers, 2014). To our knowledge this is the first report of tree regeneration on mounds created on fens where microtopography was flattened by disturbance. The influence of mounds on composition and diversity of bryophytes and vascular plants is being examined in a

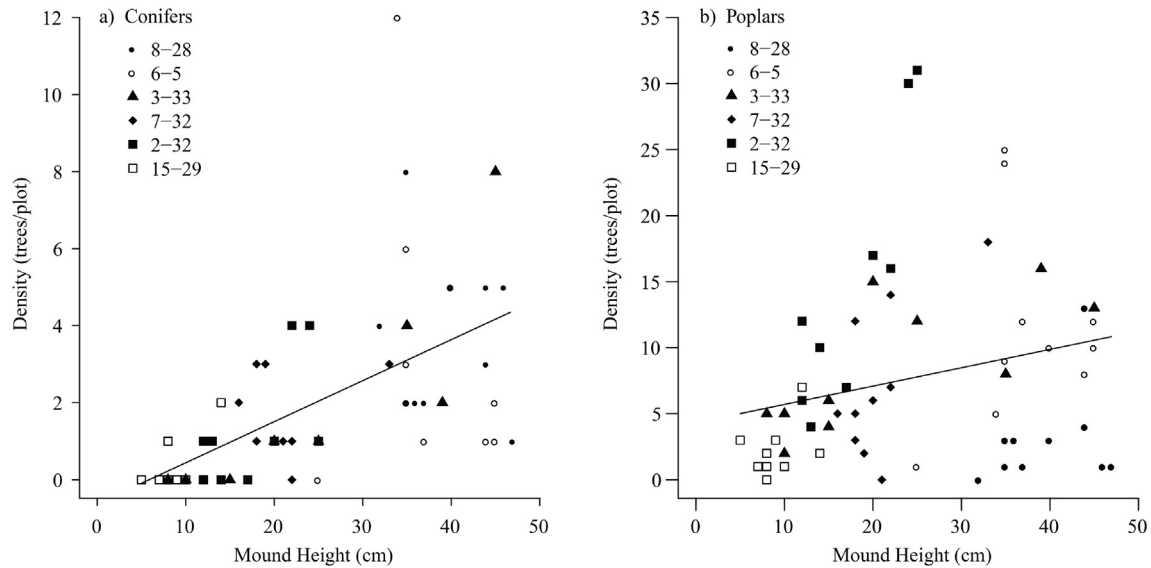


Fig. 3. Density of tree seedlings in each 10 m² plot in relation to mound height, for a) conifer species ($D = -0.595 + 0.1059H$; $r = 0.577$; $P < 0.001$; $n = 60$) and b) poplar species ($D = 4.336 + 0.1389H$; $r = 0.246$; $P < 0.06$; $n = 60$). Pad numbers are depicted by different symbols and regression statistics for individual pads are presented in [Appendix 2](#).

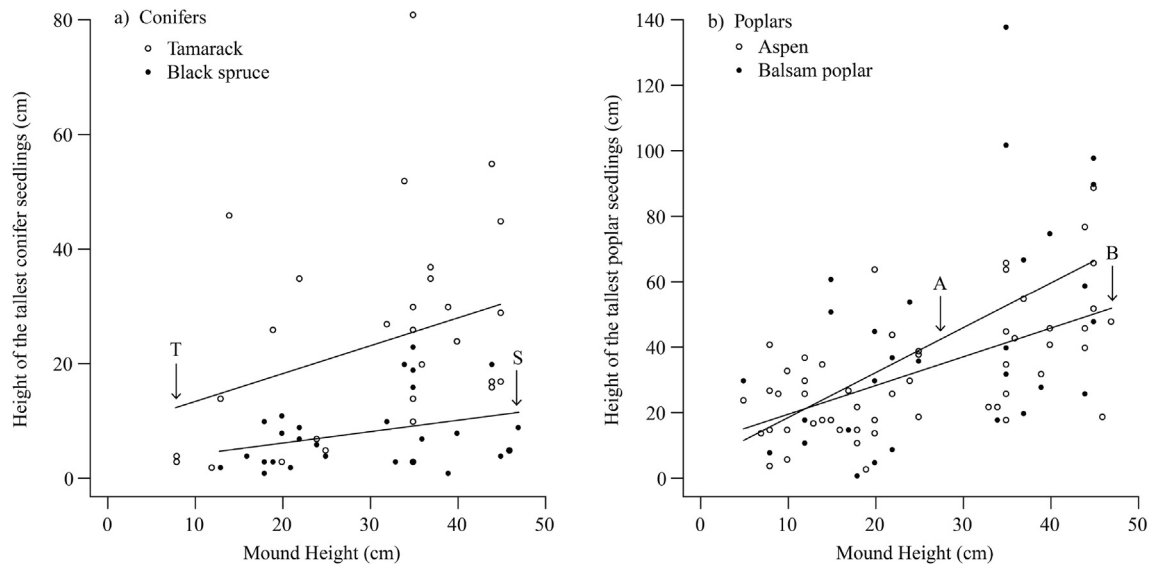


Fig. 4. Height of the tallest seedlings in relation to mound height, in 10 m² plots with at least one seedling of the focal species, four years after mounding. a) Conifer species, tamarack ($T = 8.540 + 0.4855H$; $r = 0.311$; $P < 0.09$; $n = 31$) and black spruce ($S = 2.169 + 0.1986H$; $r = 0.331$; $P < 0.089$; $n = 27$). b) Poplar species, aspen ($A = 10.947 + 0.8723H$; $r = 0.611$; $P < 0.001$; $n = 56$) and balsam poplar ($B = 4.866 + 1.3700H$; $r = 0.529$; $P < 0.003$; $n = 29$).

further study.

The height of mounds (and related variables) appears to be an important factor in the density of conifer seedlings. First, non-mounded areas had less than 20% of the tree seedling density of the mounded areas ([Fig. 2a](#)), and black spruce was absent in all unmounded areas. Second, we visited some of the pads in July 2013, following a local heavy rain event. At this time most of the non-mounded surfaces had standing water that would flood most seedlings that established within the first two years after pad abandonment. Such periodic flooding likely kills most conifer seedlings between elevated microsites; flooding reduced survival of pine seedlings in peatlands ([Pearson et al., 2011](#)). In terms of growth, however, at this early stage of recovery, the size and growth of the conifer trees was only weakly related to mound height, presumably because the full volume of peat above the water table was not yet fully occupied by the conifer roots. A larger tree, however, could only be sustained in a healthy state if its roots are able to access a large

volume of soil above the water table ([Dang and Lieffers, 1989](#)). Thus, the 1.5 m wide mounds used in this study may be an appropriate size for continued growth of these conifers, until their mound volume can be added-to by natural expansion of hummock-forming mosses ([Vitt, 1994](#); [Caners and Lieffers, 2014](#)).

A second important factor in the density of seedlings was the density and size of nearby seed sources; fewer and smaller trees would have a reduced the output of seed ([Greene and Johnson, 2000](#)). As the winged seed of conifers is expected to be blown only a short distance from its source, average density of the 10 m² plots was likely also related to the strength of seed source adjacent to each pad (some differences in stand vigour among pads are visible in [Fig. 1](#)). The regeneration success of the conifers was good despite the range of strength of seed sources among pads ([Table 1](#)).

A surprising feature of this study was the germination success of the aspen and balsam poplar on these mounded pads. There were no mature aspen or balsam poplar within these fens, so seed must

have been transported 200 m or more. Moist and well-decomposed peat substrates should be ideal germination spots for the small seeds of these poplars (Wolken et al., 2010) because of slow development of true root systems of the tiny poplar germinants in the week after germination. While these peat mounds may be ideal germination spots for poplars, it is unlikely that poplars seedlings will grow to more than a few meters in height once their roots have fully explored the peat of the mound itself. Poplars are not part of the flora of moderate-rich fens (Slack et al., 1980) and aspen has deeper rooting behaviour than either tamarack or black spruce (Strong and LaRoi, 1983).

After pad abandonment, mounds were substantial on all pads (Appendix 1), but by four years later, some of the pads had persistent and well-formed mounds (8–28 and 6–5), while others were nearly level with the surrounding pad (especially 15–29). Peat erosion has been reported in other locations where vegetation is removed and peat is exposed (e.g., Parry et al., 2014; Shuttleworth et al., 2015). When we visited 2–32 (Fig. 1d) and 15–29 in July of 2013, it was clear that most of the local erosion on these mounds had already occurred. On the four pads with lower mound height (Table 1) erosive forces were more damaging likely because the peat was thawing and fractured into smaller pieces during mound formation (Appendix 1). These four mounds had moderate to higher levels of peat humification. Erosive and shear forces will likely do more damage on heavily decomposed peat with small particle size (Tuukkanen et al., 2014) than peat with intact plant structure. We contrast this with 6–5, which also had a medium humification (H_4), but we speculate that its sustained mounds were related to the intact layer of frozen peat during mound formation (Fig. 1a). Also, the peat slabs were positioned upright, where any intact propagules were on top, providing structure to mounds. The last pad 8–28, was also formed during ongoing thaw, but its mounds remained well-formed five summers later; here mounds were formed with the least humified peat among the pads (H_2). As most of the surfaces of these peat mounds were covered by mosses or *Carex* spp. by years four and five, erosion

processes would be greatly slowed (Shuttleworth et al., 2015).

This study shows that natural regeneration could be used to establish trees provided that large and persistent mounds could be created and that seed trees are nearby. Planting of trees may also be effective but these seedlings would also need large persistent mounds until natural hummock-forming *Sphagnum* adds to the peat volume above the water table. Second, effectiveness of mounding to produce substrates for sustained tree growth will depend on the amount of mound erosion after their formation. It is likely that on sites with strong humification of the peat, the mounds will need to be formed when peat is fully frozen and peat slabs remains intact at time of lifting. Third, sites with less humified peat might allow for more persistent mounds as coarse peat would hold together better during lifting and its intact plant structures resist erosional forces. Finally, treatment of OSE pads and ice roads by mounding in the winter of exploration allows easy access for heavy equipment needed to prepare mounds.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jenvman.2017.04.002>.

Appendix 1. Pads and mounds in the first spring after formation. A) 8–28, June 26, 2012; note the blocky intact mounds. Other pads had mounds made of fractured peat: B) 3–33, June 7, 2012; C) 7–32, June 7, 2012; D) 2–32, June 7, 2012.

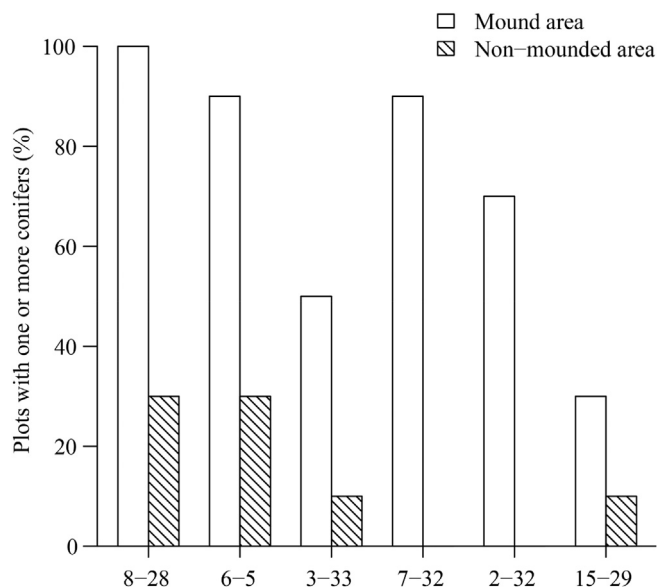


Appendix 2. Seedling density in relation to mound height for individual OSE pads for A) conifers and B) poplars.

Tables show linear regression statistics.

Location	Sample n	Slope	Intercept	r	P
A) Conifers					
8–28	10	−0.048	5.599	−0.121	0.740
6–5	10	−0.082	6.367	−0.142	0.696
3–33	10	0.169	−2.159	0.866	0.001
7–32	10	0.048	0.596	0.214	0.553
2–32	10	0.183	−1.842	0.628	0.052
15–29	10	0.143	−0.869	0.5218	0.122
B) Poplars					
8–28	10	0.228	−6.028	0.324	0.361
6–5	10	0.175	5.036	0.144	0.691
3–33	10	0.278	2.427	0.738	0.015
7–32	10	0.810	−9.557	0.672	0.033
2–32	10	1.724	−15.951	0.891	<0.001
15–29	10	0.259	−0.097	0.342	0.334

Appendix 3. Percentage of plots with at least one conifer seedling on mounded and non-mounded areas of each drilling pad. Plot size is 10 m². Pads are ranked on the basis of mean height of mounds (see Table 1). Paired *t*-test for the six pads indicates mounded and non-mounded areas differed in the percentage of plots with conifers (*P* < 0.003, *n* = 6).



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